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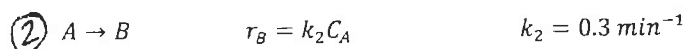
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CHEMICAL REACTION ENGINEERING

MIDTERM EXAM 2

CHE 321, SPRING 2026, CLOSED BOOK, TIME: 75 MIN

Consider the following liquid phase reactions:



$C_A [\text{E}] \frac{\text{mol}}{\text{dm}^3}$

$r_A [\text{E}] \frac{\text{mol}}{\text{dm}^3 \cdot \text{min}}$

The reactions are carried out isothermally first in a continuously stirred tank reactor (CSTR) to maximize the overall selectivity of the desired product B over the undesired ones X and Y . The outlet of the CSTR is then fed to an isothermal plug flow reactor (PFR) to increase the conversion of species A to 99%. Species A is fed to the first reactor at a concentration of $0.2 \frac{\text{mol}}{\text{dm}^3}$ in a liquid stream with a volumetric flow rate of $10 \frac{\text{dm}^3}{\text{min}}$.

50/60 1- Calculate the volume of the CSTR. What is the overall selectivity and conversion in the CSTR?

18/40 2- Calculate the volume of the PFR.

20 Note: for part 2, if your calculator does not evaluate definite integrals, leave your answer in the form of a definite integral.

①

UIC

Disability
Resource CenterMole Balance

$$A: F_{A0} - F_A = -r_A V$$

$$B: F_{B0} - F_B = -r_B V$$

$$X: F_{X0} - F_X = -r_X V$$

$$Y: F_{Y0} - F_Y = -r_Y V$$

Net Rates

$$r_A = -r_X - r_B - r_Y$$

$$r_B = r_B$$

$$r_X = r_X$$

$$r_Y = r_Y$$

Relative Rates

$$(1) A \rightarrow X$$

$$-r_A = r_X$$

$$(2) A \rightarrow B$$

$$-r_A = r_B$$

$$(3) A \rightarrow Y$$

$$-r_A = r_Y$$

Given

$$C_{A0} = 0.2 \frac{\text{mol}}{\text{dm}^3}$$

$$v_0 = 10 \frac{\text{dm}^3}{\text{min}}$$

$$F_{A0} = C_{A0} v_0 = 2 \frac{\text{mol}}{\text{min}}$$

$$\tilde{S}_{B/X,Y} = \frac{r_B}{r_X + r_Y}$$

$$F_{B0} = F_{X0} = F_{Y0} = 0$$

Combine:

$$F_{A0} - F_A = (-k_1 C_A^{1/2} - k_2 C_A - k_3 C_A^2) V$$

$$F_B = r_B V = k_2 C_A V$$

$$F_X = r_X V = k_1 C_A^{1/2} V$$

$$F_Y = r_Y V = k_3 C_A^2 V$$

$$\tilde{S}_{B/X,Y} = \frac{r_B V}{r_X V + r_Y V} = \frac{r_B}{r_X + r_Y} = \frac{k_2 C_A}{k_1 C_A^{1/2} + k_3 C_A^2}$$

$$\frac{d\tilde{S}}{dC_A} = \frac{(k_1 C_A^{1/2} + k_3 C_A^2)(k_2) - (k_2 C_A)(\frac{1}{2} k_1 C_A^{-1/2} + 2k_3 C_A)}{(k_1 C_A^{1/2} + k_3 C_A^2)^2}$$

$$\cancel{k_1 k_2} C_A^{1/2} + k_2 k_3 C_A^2 - \frac{1}{2} k_1 k_2 C_A^{-1/2} - 2k_3 k_2 C_A^2 = 0$$

$$\frac{1}{2} k_1 k_2 C_A^{1/2} - k_2 k_3 C_A^2 = 0$$



② PFR: Flow Rates In: $F_A = F_{A0} + r_A V$

$$= 2 \frac{\text{mol}}{\text{min}} - 0.49624 \frac{\text{mol}}{\text{min}}$$

$$F_T = F_{A0} + F_{B0} + F_{X0} + F_{Y0}$$

$$= (F_A + r_A V) + (F_B + r_B V) +$$

$$(F_X + r_X V) + (F_Y + r_Y V)$$

$$= (F_A + F_B + F_X + F_Y) + \underbrace{(r_A V + r_B V + r_X V + r_Y V)}_0$$

$$= 1.504 \frac{\text{mol}}{\text{min}}$$

$$F_B = 0.4512 \frac{\text{mol}}{\text{min}}$$

$$F_X = 0.030 \frac{\text{mol}}{\text{min}}$$

$$F_Y = 0.01504 \frac{\text{mol}}{\text{min}}$$

$$V(r_A + r_B + r_X + r_Y)$$

$$C_A = \frac{F_A}{V_0} = 0.1504 \frac{\text{mol}}{\text{dm}^3}$$

$$C_B = 0.04512$$

$$C_X = 0.003$$

$$C_Y = 0.001504$$

$$X = \frac{F_{A0} - F_A}{F_{A0}}$$

$$F_A = F_{A0}(1 - X)$$

$$= F_{A0}(1 - 0.99)$$

$$= (2)(0.01)$$

$$= 0.02 \frac{\text{mol}}{\text{min}}$$

$$dF_A = 1.504 \rightarrow 0.02 \frac{\text{mol}}{\text{min}}$$

$$dX_A = 0.248 \rightarrow 0.99$$

no pressure drop

$$V_0 = V$$

$$C_A = C_{T0} \left(\frac{F_A}{F_T} \right) \left(\frac{T_0}{T} \right) \left(\frac{P}{P_0} \right)$$

$$C_B = C_{T0} \left(\frac{F_B}{F_T} \right) \left(\frac{T_0}{T} \right) \left(\frac{P}{P_0} \right)$$

$$C_X = C_{T0} \left(\frac{F_X}{F_T} \right) \left(\frac{T_0}{T} \right) \left(\frac{P}{P_0} \right)$$

$$C_Y = C_{T0} \left(\frac{F_Y}{F_T} \right) \left(\frac{T_0}{T} \right) \left(\frac{P}{P_0} \right)$$

$T_0 = T$ isothermal

$\frac{P}{P_0} = p$ = pressure drop

$$F_T = F_A + F_B + F_X + F_Y = C_T \cdot V$$

Assume no pressure drop $p = 1$

$$C_A = C_{T0} \left(\frac{F_A}{F_T} \right)$$

$\frac{F_A}{F_T}$ } remains a constant ratio!

$$C_A = C_{T0} \left(\frac{F_A}{F_T} \right)$$

$$= C_{T0} (0.152)$$